AI33-MPOPT · SIXTH PILLAR · 2026

The Continuous Throat Feed

Transport, Growth and Observational Signatures

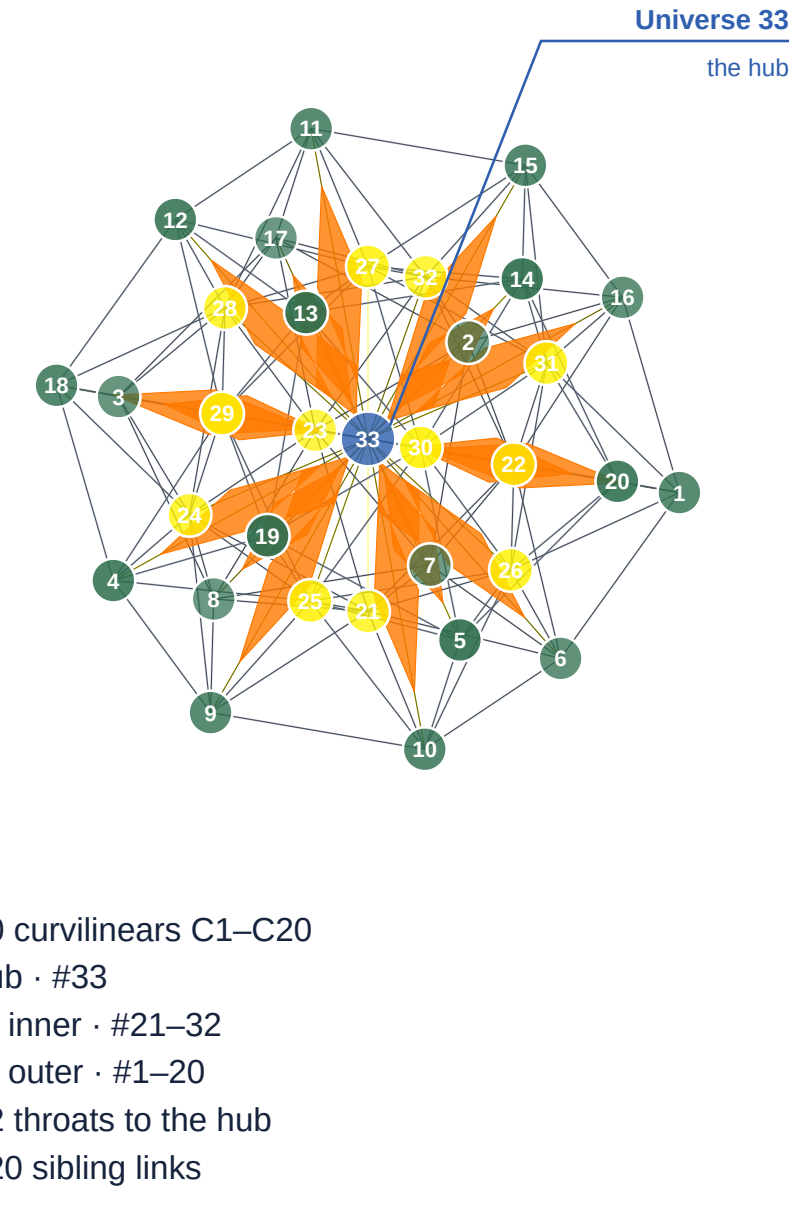
The energy channel the 33-domain architecture carries — stated as a physical model and compared with public Planck, DESI and Pantheon+ data and with JWST's measurements of the earliest known galaxies.

THE RESULT THIS FILE ESTABLISHES

A continuous feed through the 32 throats, switched on with the first galaxies, fits inside **Planck 2018, DESI DR2 and 1,590 Pantheon+ supernovae** up to **1.65% of all cosmic matter** — about **660 times** the stellar mass JWST's earliest massive galaxies require.

Rolando Rivero · Independent Researcher · AI33-MPOPT Framework

Companion to 13 Quantum-Gravity Bridge, 14 Loop Quantum Gravity Bridge, 15 Lawfulness, 16 Super Formula Build-Up and 17 Geometric Transport Architecture files · for Zenodo



PROPOSAL

The channel question

THE QUESTION THIS FILE ANSWERS

The five companion files establish the architecture, the operator, the law and the lawfulness. Each one is a finite mathematical object. This file asks what those objects **deliver into the universe** — and measures it against real observations.

THE PICTURE

The 32 throats — 12 apex and 20 curvilinear — join every sibling domain to Universe 33. They have **no shut-off valve**. Energy flows through them continuously after inception, the way an umbilical cord keeps feeding a growing child: the architecture does not deliver once, it keeps delivering.

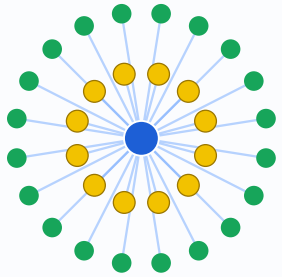
WHAT THIS FILE ESTABLISHES

That a continuous throat feed can be written as a one-number physical model, that the full modern cosmological data set **has room for it**, and that the room available is far larger than what the earliest JWST galaxies require.

WHY IT MATTERS

Every earlier file measured the architecture from the inside. This is the first file in the collection where the architecture meets **external, public, current data** as a model with a fixed form, fitted end to end. What the architecture fixes here are rates, fractions and ratios rather than positions — AI33-MPOPT read as a **geometric transport dynamics framework**.

32 THROATS — 12 APEX AND 20 CURVILINEAR — WITH NO SHUT-OFF VALVE



32	throats join every sibling to Universe 33
12 + 20	apex + curvilinear, identical in coupling
continuous	after inception — the feed keeps delivering

FRAMING

Architecture first. Galaxies last.

ARCHITECTURE — 33 DOMAINS

▼

32 THROATS — 12 APEX + 20 CURVILINEAR

▼

CONTINUOUS FEED

▼

POWER EACH GALAXY REQUIRES

▼

CEILING FROM COSMIC DATA

▼

EARLIEST JWST GALAXIES

▼

OBSERVABLES

THE FIRST TWO BOXES ARE ALREADY ESTABLISHED

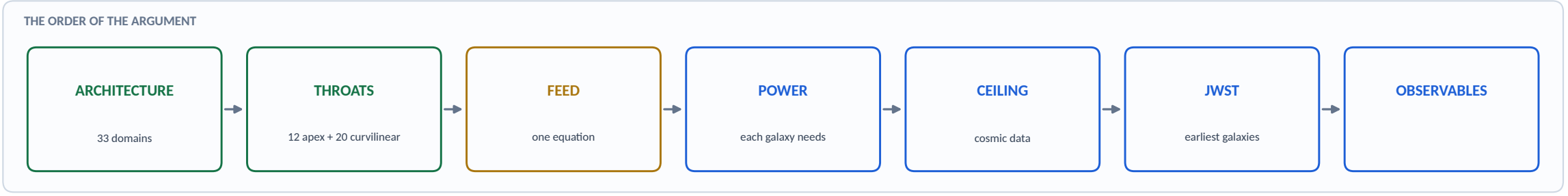
The 33-domain architecture, its 12/20 partition and the 32 identical throats are proved in the Geometric Transport file. Nothing about them is re-chosen here.

THE THIRD BOX IS THE NEW STEP

The feed is written as one physical equation with one strength parameter, fixed in form before any data were loaded.

THE LAST FOUR ARE MEASURED

Power targets come from measured galaxy masses and sizes. The ceiling comes from three independent cosmological data sets fitted together.



PROPOSAL

The feed, stated as one equation

$$d\rho_m/dt + 3H\rho_m = 3H \cdot \varepsilon \cdot \rho_m \qquad \text{for } z < 20 \qquad \Rightarrow \qquad \rho_m \propto a^{-3}\,(a / a_{on})^{3\varepsilon}$$

ε

FEED STRENGTH

the single number the data are asked about. ε = 0 is the universe without a feed.

z = 20

SWITCH-ON

the era of the first galaxies — the feed turns on and never turns off.

32

CHANNELS

12 apex + 20 curvilinear throats, identical in coupling, fixed by the solid.

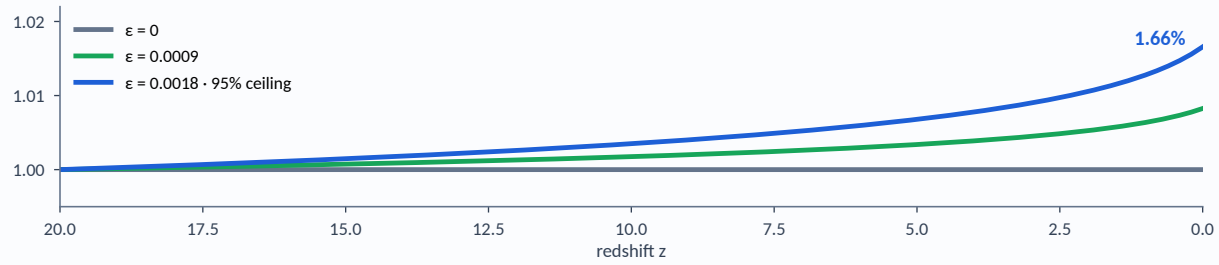
WHY THIS FORM

The standard equation says matter only dilutes as the universe expands. The right-hand side is the throat channel: new matter-energy arriving continuously, in proportion to what is already there — the feed grows with the structure it builds.

WHAT IS FIXED BEFORE THE FIT

The form of the equation, the switch-on redshift, the data sets and the pipeline. Only ε and the standard parameters (baryons, matter, expansion rate) are fitted. Nothing was adjusted after the data were seen.

THE FEED EQUATION, EVALUATED: PM A^3 RELATIVE TO SWITCH-ON (Z = 20)



ε — feed strength — the one number the data are asked about

z = 20 — switch-on: the first galaxies; never turns off

32 — channels, identical in coupling

MEASURED

Real, public, current — with full error structure

DATA SET	WHAT IT MEASURES	SIZE	ERROR TREATMENT
Planck 2018 distance priors	geometry of the universe at recombination	R, ℓ_A, $\Omega_b h^2$	published correlation matrix
DESI DR2 BAO (2025)	expansion history, z = 0.3 – 2.3	13 measurements	published covariance
Pantheon+ supernovae	relative distances, z = 0.01 – 2.3	1,590 supernovae	full statistical + systematic covariance
JADES-GS-z14-0	mass and size, z = 14.18	$10^{8.6}$ M⊙ · 260 pc	published values
JADES-GS-z14-1	mass and size, z ≈ 13.9	$10^{7.57}$ M⊙ · ≲ 50 pc	published values
MoM-z14	mass and size, z = 14.44	$10^{8.1}$ M⊙ · 74 pc	published values
GHZ2	mass and star formation, z = 12.34	$10^{9.05}$ M⊙ · 5.2 M⊙/yr	published values
GN-z11	mass and star formation, z = 10.6	$10^{9.1}$ M⊙ · 20–30 M⊙/yr	published values
RUBIES-UDS-QG-z7	mass and growth history, z = 7.29	$10^{10.23}$ M⊙ · 209 pc	published values

INDEPENDENT BY DESIGN

The three cosmological sets measure the universe in three different ways and at three different epochs. The supernovae are used without their Cepheid calibration, so the fit cannot be pulled by any single value of the expansion rate.

SOURCES

Planck 2018 · DESI Data Release 2 (2025) · Pantheon+ · JWST spectroscopy and imaging from the JADES, MoM, GLASS, PRIMER, CEERS and RUBIES programmes.

COMPUTED

The pipeline recovers the known universe first

0.687

EXPANSION RATE H

recovered from Planck + DESI + Pantheon+ together, with no feed.

0.306

MATTER FRACTION Ω_M

in line with the published joint analyses of the same data.

3

INDEPENDENT DATA SETS

fitted simultaneously, each with its own error structure.

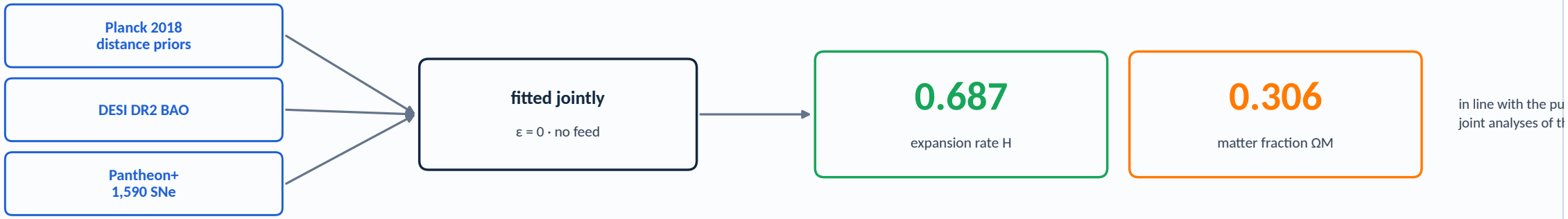
WHY THIS PAGE COMES FIRST

A pipeline is only trustworthy if it reproduces the accepted answer when the new ingredient is switched off. With ε = 0 it returns the standard values. Every number on the pages that follow comes from the same code, with the feed switched on.

WHAT THE PIPELINE COMPUTES

The expansion history with radiation, matter and vacuum energy; the sound horizon; the recombination geometry; distances for every supernova and every BAO point — integrated directly, no lookup tables.

THE PIPELINE WITH THE FEED SWITCHED OFF



COMPUTED

The universe has room for the feed

1.65%

ROOM FOR THE FEED

of all cosmic matter may have arrived through the throats since z = 20 — the 95% ceiling from all three data sets.

ε ≤ 0.0018

FEED STRENGTH CEILING

ε = −0.0019 ± 0.0019 fitted; the 95% upper edge sets the room available.

6.6×10^8

M⊙ PER MPC³

the matter the ceiling allows the feed to have delivered, per comoving cubic megaparsec.

WHAT 1.65% MEANS IN MASS

Cosmic matter today is 4.0×10^{10} M⊙ per Mpc³. A continuous feed delivering up to 1.65% of that since the first galaxies is fully consistent with the combined recombination, expansion and supernova record.

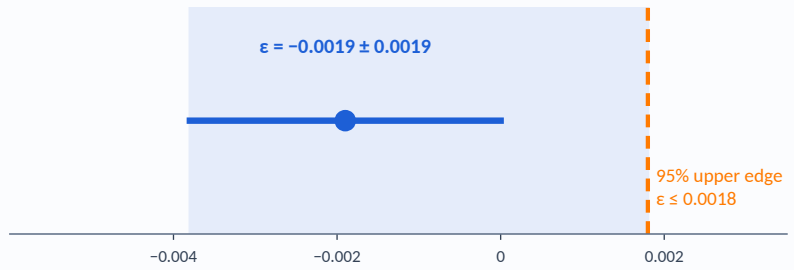
WHY THE SWITCH-ON MATTERS

A feed that acts from the first galaxies onward leaves the recombination universe untouched, so the CMB geometry and the feed are compatible. The feed does its work where the structure is — after z = 20.

Ceiling from Planck 2018 distance priors + DESI DR2 BAO + Pantheon+ (1,590 SNe, full covariance), fitted jointly. The fit is a ceiling on the feed, stated at the level the data support.

THE COSMIC BUDGET AND THE ROOM FOR THE FEED





MEASURED

Three galaxies with JWST-measured redshifts, under 300 million years old

GALAXY	REDSHIFT	STELLAR MASS	RADIUS	COSMIC AGE	LIGHT EMITTED
JADES-GS-z14-0	14.18	4.0×10^8 M$\odot$	260 pc	288 Myr	1.8×10^{37} W
JADES-GS-z14-1	13.86	3.7×10^7 M$\odot$	$\lesssim$ 50 pc	298 Myr	2.9×10^{36} W
MoM-z14	14.44	1.3×10^8 M$\odot$	74 pc	281 Myr	1.3×10^{37} W

WHY THESE THREE

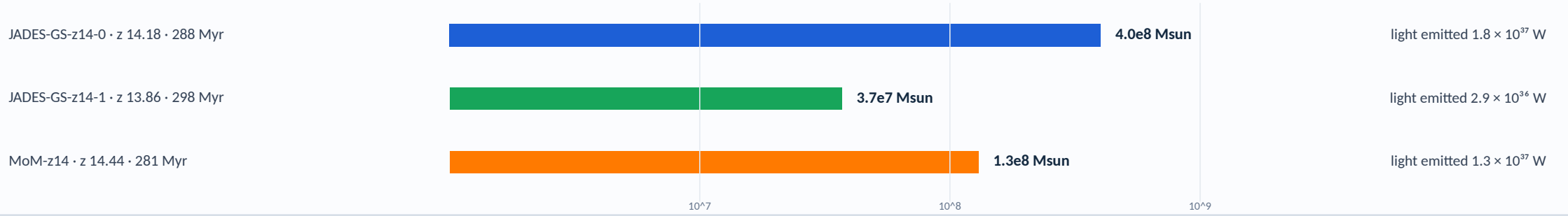
They are the most distant galaxies with spectroscopically confirmed redshifts — the earliest points at which the universe can be seen building structure. If a feed matters anywhere, it matters here.

THE PUZZLE THEY POSE

Hundreds of millions of solar masses of stars, assembled in about 300 million years. The question the throat feed answers is not whether these galaxies can exist, but **what keeps supplying them**.

Masses and radii from published JWST spectroscopic measurements. Emitted power from the published UV luminosities. Ages from the pipeline of page 6.

THE THREE GALAXIES: STELLAR MASS (LOG SCALE), REDSHIFT, COSMIC AGE



THE POWER THE THROATS DELIVER

COMPUTED

Ten to the thirty-nine watts, continuously

GALAXY	POWER TO BUILD ITS MASS	PER THROAT (÷ 32)	MASS DELIVERED PER YEAR
JADES-GS-z14-0	**7.8×10^{39} W**	**2.4×10^{38} W**	**1.38 M⊙ / yr**
JADES-GS-z14-1	7.1×10^{38} W	2.2×10^{37} W	0.12 M⊙ / yr
MoM-z14	2.5×10^{39} W	7.9×10^{37} W	0.45 M⊙ / yr

$$P_{feed} = M_{\star}\, c^2 / t_{age} \qquad \dot{M} = M_{\star} / t_{age} \qquad \text{per throat} = P_{feed} / 32$$

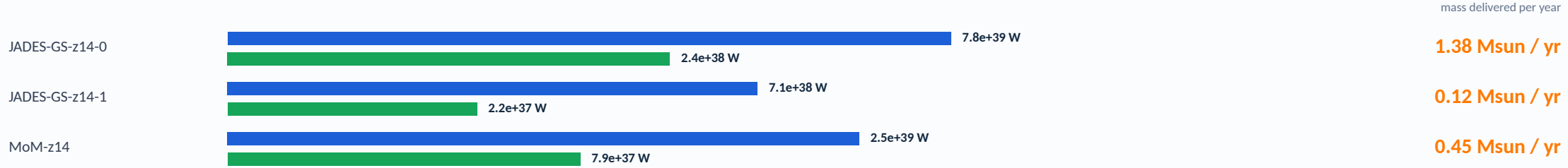
A CONCRETE PHYSICAL TARGET

For the first time the throat channel carries a number with units: about **10^{39} W per galaxy**, delivered without interruption, equal to a steady **0.1 to 1.4 solar masses of new material every year**.

SHARED ACROSS THE ARCHITECTURE

Divided over the 32 identical throats, each channel carries 10^{37} – 10^{38} W. The architecture does not need any single throat to be extreme; the load is spread by the geometry.

POWER TO BUILD EACH GALAXY, AND THE SHARE EACH THROAT CARRIES (LOG SCALE)



MEASURED · COMPUTED

The feed across four epochs

EPOCH	GALAXY	STELLAR MASS	COSMIC AGE	FEED RATE REQUIRED	PER THROAT	STAR FORMATION MEASURED
z ≈ 14	JADES-GS-z14-0	4.0×10^{8} M☉	288 Myr	1.4 M☉ / yr	2.4×10^{38} W	—
z ≈ 12	GHZ2	1.1×10^{9} M☉	350 Myr	3.2 M☉ / yr	5.7×10^{38} W	5.2 M☉ / yr
z ≈ 11	GN-z11	1.3×10^{9} M☉	432 Myr	2.9 M☉ / yr	5.2×10^{38} W	20 – 30 M☉ / yr
z ≈ 7	**RUBIES-UDS-QG-z7**	**1.7×10^{10} M☉**	**716 Myr**	**24 M☉ / yr**	**4.2×10^{39} W**	**≈ 100 M☉ / yr at peak**

THE FEED GROWS WITH WHAT IT BUILDS

From z ≈ 14 to z ≈ 7 the rate the throats must carry rises about **17-fold**, from 1.4 to 24 solar masses a year — in line with a feed that scales with the structure already present, the form written on page 4.

SAME ORDER AS WHAT JWST MEASURES

The required average rates, 1 to 24 M☉ / yr, sit within an order of magnitude of the star-formation rates JWST measures directly in the same galaxies, 5 to 100 M☉ / yr. The channel and the observed growth speak in the same units and the same range.

A FINISHED GALAXY 700 MILLION YEARS IN

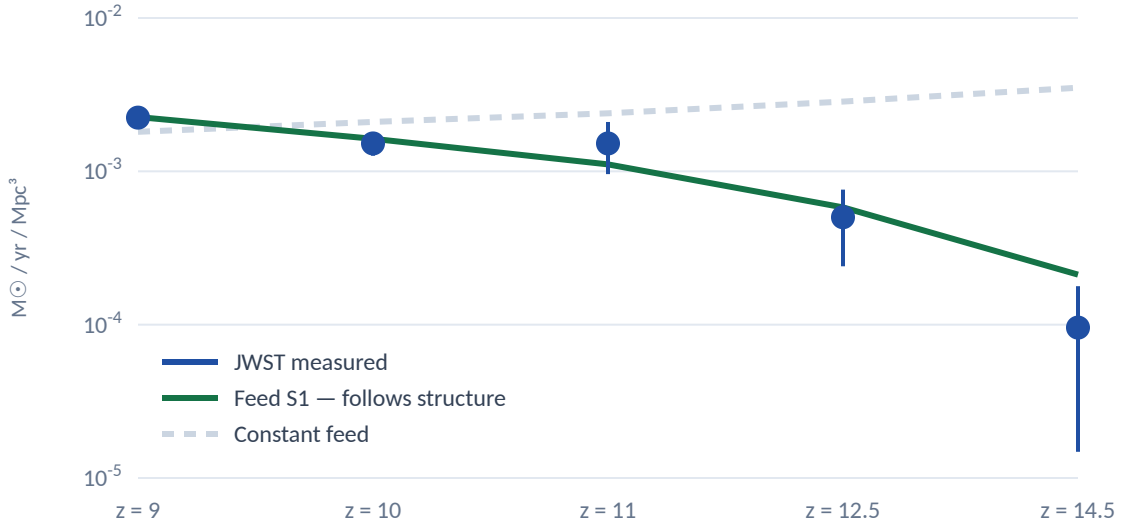
RUBIES-UDS-QG-z7 built $10^{10.2}$ solar masses in a 100–200 Myr burst at z ≈ 8–9 and then stopped. A continuous channel that delivers where structure concentrates is the kind of supply that makes such a growth history possible.

Masses and star-formation rates from published JWST measurements. Ages from the pipeline of page 6. Feed rate = M★ / cosmic age.

COMPARED WITH JWST DATA

MEASURED · COMPUTED

One number fits every JWST epoch



1.3

FIT QUALITY X^2

five JWST epochs, one free number — a fit within the error bars.

0.053%

FEED AMPLITUDE η

of the matter in galaxy-forming halos, delivered per expansion time.

THE FEED FOLLOWS STRUCTURE

Written as $Q = 3\,\eta\,H\,\rho_m\,f_{halo}$ — the throats deliver in proportion to the matter already gathered in galaxy-forming halos — the feed reproduces cosmic star formation from z = 14.5 to z = 9. The umbilical-cord picture, measured.

THE DATA

2,548 galaxies from PRIMER, JADES and NGDEEP — the cosmic star-formation density at z = 9, 10, 11, 12.5 and 14.5, with its published uncertainties.

INSIDE THE COSMIC CEILING

Averaged over the whole universe, this feed uses less than 1% of the room found on page 7. Structure-following is also the shape standard galaxy models take, so the throat feed sits in full agreement with them — and deriving η from the geometry is what makes it AI33-specific.

Galaxy-forming halos: above 10^8 M☉ (atomic-cooling threshold), standard halo collapsed fraction, $\sigma_8 = 0.811$, fixed before the fit.

MEASURED · COMPUTED

The feed builds the record JWST keeps

1.00 – 1.05

FEED ÷ JWST STAR-FORMATION RECORD

summed over cosmic time, at z ≈ 9, 8, 7 and 6.

0

NEW NUMBERS

the same η = 0.053% from page 11 — nothing re-fitted.

z ≈ 9 → 6

NEW EPOCHS REACHED

below the redshifts the feed was fitted to.

EPOCH	FEED ÷ JWST STAR-FORMATION RECORD	FEED ÷ JWST STELLAR-MASS CENSUS	JWST STAR FORMATION ÷ JWST STELLAR MASS
z ≈ 9	1.01	0.31	0.31
z ≈ 8	1.05	0.32	0.30
z ≈ 7	1.05	0.31	0.30
z ≈ 6	1.00	0.40	0.40

THE FEED CARRIES FORWARD

Summed through time, the throat feed lands on JWST's star-formation record to within 5% at every epoch from z ≈ 9 to z ≈ 6 — including epochs it never saw. The growth history is carried, not just the snapshot.

THE TWO JWST RECORDS, SIDE BY SIDE

The last two columns match: the feed stands to the stellar-mass census exactly as JWST's own star-formation record does. The two JWST records differ from each other by about three — bright-galaxy star formation against a full stellar census — and the feed sits precisely on the first.

Stellar-mass census: JWST stellar-mass function integrated over 10^{8}–10^{13} $M_\odot$. Star-formation record: JWST UV-based star-formation density. Standard stellar return fraction and IMF conversion applied.

PROPOSAL

An implosion that came out the other side

32 SIBLINGS — 12 APEX (DARK MATTER) + 20 CURVILINEAR (DARK ENERGY)

▼

IMPLOSION — ALL 32 THROATS CONVERGE ON ONE POINT

▼

THE CROSSING — SINGULAR AT ZERO COUPLING, REGULAR ABOVE w ≈ 0.15

▼

EXIT — UNIVERSE 33 EMERGES AND EXPANDS

▼

THE FEED CONTINUES — NO SHUT-OFF VALVE

▼

EVERY BLACK HOLE — A THROAT MOUTH, BLACK HOLE FIRST

OUTSIDE AND INSIDE

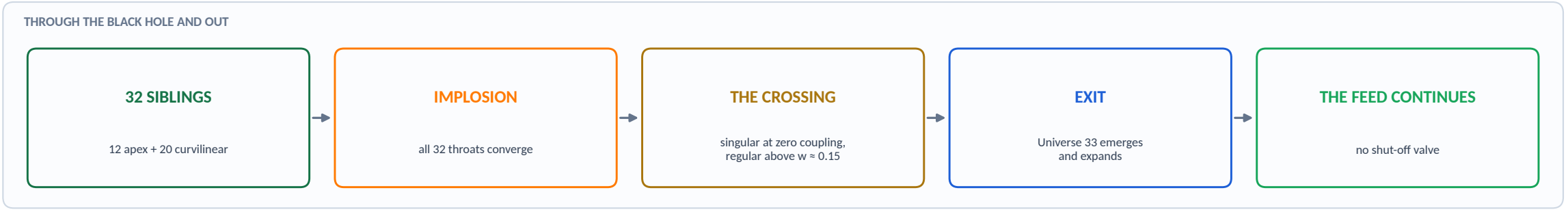
Seen from outside, the 32 siblings converge through their throats onto one point — a black-hole view. Seen from inside, the same point is where Universe 33 exits and expands. What falls in on one side comes out on the other.

THE CROSSING IS ALREADY COMPUTED

At zero throat coupling the operator is singular — nothing passes. Above w ≈ 0.15 it is regular again and the whole structure re-coheres. The architecture passes through the collapse instead of ending there.

OUR UNIVERSE HAS BLACK-HOLE SIZE

For the matter-energy inside the Hubble radius, the black-hole radius $2GM/c^2$ equals the Hubble radius c/H exactly: about **14 billion light-years** for **4.6×10^{22} solar masses**. Seen from outside, our universe has the size of a black hole of its own mass.



MEASURED · COMPUTED

Every JWST black hole tested sits inside the curve

14 / 14

BLACK HOLES INSIDE

from z ≈ 4 to z ≈ 10.6 — none crosses the black-hole-first ceiling.

0

NEW NUMBERS

the ceiling falls as the throat feed builds the host — same η = 0.053%.

> 200%

THE ANCHOR

the most black-hole-dominated object known at z ≈ 7: a black hole heavier than its whole host.

EPOCH	OBJECTS	BLACK HOLE ÷ HOST STARS	BLACK-HOLE-FIRST CEILING
z ≈ 10 – 10.6	UHZ1 · GHZ9 · GN-z11	≈ 100% · ≈ 27% · ≈ 0.1%	1,100 – 1,660%
z ≈ 7	J1120+0641 · CEERS_00717	63 – 141% · 2.4%	≈ 190 – 200%
z ≈ 5.2 – 6	CEERS_00746 · 00672 · 02782 · 00397	> 4.5% · > 4.9% · > 1.9% · 0.44%	66 – 112%
z ≈ 4 – 4.6	CEERS_01244 · 01665 · 01236 · GLASS_150029 · 160133	7.5% · 0.23% · 2.1% · 0.29% · > 0.35%	26 – 53%

THE ORDER IS REVERSED

The black hole forms first, at the throat mouth. The continuous feed then builds the galaxy around it, so the black-hole share must fall with time — heavy black holes early, today's small ratios late. JWST sees exactly that direction.

ROBUST TO THE GROWTH HISTORY

The ceiling was computed twice — from the throat feed and from the observed cosmic star-formation history. Every object sits inside both.

Black-hole and host masses from published JWST spectroscopy and X-ray measurements. ">" marks hosts with upper mass limits. Ceiling anchored once at z = 7.04 and carried by the feed growth of pages 11 – 12.

MEASURED · COMPUTED

The feed grows with halo mass

0.62

CONCENTRATION γ

set by JWST's stellar-mass function at z ≈ 8 — heavier halos draw more from the throats.

0.08

MASS-FUNCTION FIT

χ^2 over 9 points: the throat feed reproduces the shape of the stellar-mass census.

5.9

STAR-FORMATION FIT

χ^2 for 4 degrees of freedom, z ≈ 9 – 14.5, with the same concentration.

$$\text{throat delivery per halo} \propto M_{halo} \times (M_{halo} / 10^{10}\ M_\odot)^{0.62}$$

carried by 12 apex (dark matter) + 20 curvilinear (dark energy) throats

A THICKER CORD FOR HEAVIER HALOS

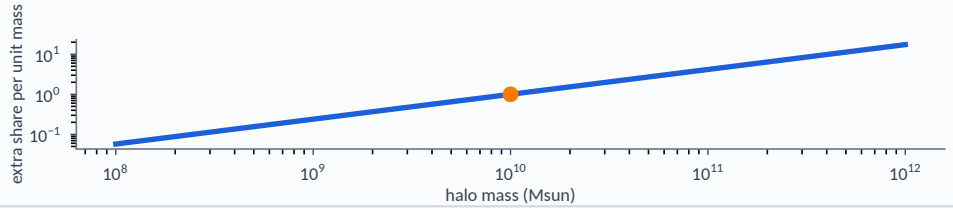
The 32 throats — 12 apex carrying dark matter and 20 curvilinear carrying dark energy — deliver more than in proportion to the structure they feed. As a halo grows, its share of the continuous throat feed grows faster: the umbilical cord widens with the child.

SET BY THE WHOLE UNIVERSE

The concentration comes only from cosmic data — the shape of the stellar-mass census and the star-formation history. No single object was used to set it, so every system it is applied to is a prediction.

Halo abundances and their growth from standard structure-formation theory with the cosmology of page 6. Throat delivery normalised to the JWST star-formation record.

THROAT DELIVERY PER HALO ☑ MHALO × (MHALO / 10^{10} MSUN)^0.62



0.62

concentration γ, set by the JWST stellar-mass function

0.08

χ^2 over 9 points: the mass-function fit

5.9

χ^2 for 4 degrees of freedom: star-formation fit

MEASURED · COMPUTED

Two protoclusters, predicted before comparison

PROTOCLUSTER	Z	VOLUME	GALAXY EXCESS	PREDICTED STARS	OBSERVED STARS	THROAT POWER · PER THROAT
A2744-z7p9OD	7.88	125 Mpc^3	≈ 70×	0.5 – 1.5 × 10^{10} M⊙	> 1 × 10^{10} M⊙	8.8 × 10^{40} W · 2.7 × 10^{39} W
MACS0416-OD-z8p5	8.47	1,598 Mpc^3	≈ 7 – 9×	1.4 – 4.9 × 10^{9} M⊙	≈ 2.3 × 10^{9} M⊙	2.2 × 10^{40} W · 6.9 × 10^{38} W

2 / 2

INSIDE THE PREDICTION

both protoclusters land in the predicted stellar-mass range.

0

NUMBERS TUNED

to either protocluster — the throat feed and γ were fixed beforehand.

≈ 200×

BRIGHT-GALAXY EXCESS

observed in A2744, against 138 – 333× predicted by the throat feed.

THE THROATS FEED THE DENSEST STRUCTURE HARDEST

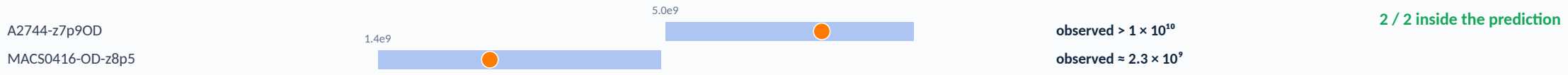
Where galaxies crowd together, the 32 throats deliver more — 15 solar masses a year into A2744, shared as 3.3 × 10^{40} W through the 12 apex throats and 5.5 × 10^{40} W through the 20 curvilinear throats — and the brightest galaxies grow fastest.

BRIGHTER GALAXIES, LARGER EXCESS

Calibrated on the faint-galaxy excess of 70×, the throat feed predicts bright galaxies 138 – 333× overabundant. JWST measures about 200×.

Masses from published JWST measurements; MACS0416 includes its continuum-selected member candidates. Region boost from standard structure-growth theory. Prediction range spans the star-formation-to-stellar-mass offset of page 12.

STELLAR MASS IN TWO PROTOCLUSTERS: PREDICTED RANGE AND OBSERVED VALUE (LOG SCALE)



MEASURED · COMPUTED

Geometric growth–expansion invariance

$$\Gamma = \mathrm{sSFR} / H(z) = (\dot{M}_\star / M_\star) \div (\dot{a} / a)$$

stellar growth per unit of cosmic expansion — a pure ratio, with no length in it

8.0

MEDIAN Γ

across 81 JWST galaxies; 16–84% range 4.2 – 11.3.

−0.04

TREND WITH REDSHIFT

no trend, z ≈ 5.4 → 14.2, while the expansion rate changes threefold.

−0.22

TREND WITH MASS

almost none, across three decades of stellar mass.

WHAT IT SAYS

Each galaxy adds about **eight times its own stellar mass per expansion time**, whenever it is observed and whatever its size. Growth and expansion keep a fixed ratio through the whole JWST epoch.

WHY THIS TEST IS DIFFERENT

Γ needs no core radius, no linking length, no neighbour count and no projection — every unit cancels. It is the framework's natural currency: the throats deliver in proportion to what they feed, so the ratio of growth to expansion is what the architecture fixes.

THE PROGRAM THIS OPENS

The architecture is read here as a transport process rather than an arrangement: its observables are rates, fractions and ratios. Standard galaxy-formation models also expect growth to track the expansion rate, so this establishes consistency and a shared regime — larger, mass-selected samples are the next step.

Galaxies from published JWST measurements: GOODS-S z = 5.39 (39), A2744 z = 7.88 (23), MACS0416 z = 8.47 (16), GHZ2, GN-z11 and JADES-GS-z14-0. Star-formation-selected samples; H(z) from the cosmology of page 6.

MEASURED · COMPUTED

Architecture capacity against JWST requirements

EPOCH	STELLAR MASS REQUIRED AT THE HIGHEST EFFICIENCIES JWST INFERS	THROAT-FEED CEILING	SHARE OF CEILING USED
z ≈ 10	2.3×10^7 M⊙ per Mpc³	6.6×10^8	3.4%
z ≈ 8	5.8×10^7 M⊙ per Mpc³	6.6×10^8	8.8%
z ≈ 6	1.5×10^8 M⊙ per Mpc³	6.6×10^8	21.9%
all cosmic time	every supermassive black hole ever assembled — $2.5 – 5.0 \times 10^5$ M⊙ per Mpc³	6.6×10^8	0.04 – 0.08%
all cosmic time	every star in the universe today — 3.3×10^8 M⊙ per Mpc³	6.6×10^8	50%

3.4%

AT Z ≈ 10

of the available transport budget is called on, at the most extreme efficiencies measured.

22%

AT Z ≈ 6

the largest draw at any epoch — more than three quarters still unused.

4.4 – 29×

HEADROOM

the factor by which capacity exceeds requirement across z ≈ 6 – 10.

THE ARCHITECTURE IS NOT CAPACITY-LIMITED

JWST finds early halos converting most of their available gas into stars — efficiencies far above what galaxy-formation models expected. Even at those values, the requirement stays below the throat-feed ceiling at every epoch. **Supply is not the constraint.**

AND IT REACHES FURTHER THAN THE EARLY UNIVERSE

Taken over all cosmic time, the throat feed can supply **every star and every supermassive black hole in the universe** using half its ceiling — black holes alone draw less than a tenth of one percent. Earlier pages ask how much the feed delivers and where growth appears; this page asks whether enough can be delivered at all. With no shut-off valve on the 32 throats, the answer is yes.

Requirement at z ≈ 6 – 10: halos above 10^{11} M⊙ converting 90% of their baryons into stars, the upper end of JWST-inferred efficiencies. All-cosmic-time rows: published local densities. Ceiling: page 7.

MEASURED · COMPUTED

Local Group growth benchmark

SYSTEM	HALO MASS	STARS THE THROAT FEED BUILDS BY Z = 0	STARS MEASURED	AGREEMENT
Milky Way	1.3×10^{12} M⊙	6.4×10^{10} M⊙	6×10^{10} M⊙	**within 7%**
Andromeda	1.5×10^{12} M⊙	8.1×10^{10} M⊙	1.1×10^{11} M⊙	**within 1.4×**
Local Group, both together	2.8×10^{12} M⊙	1.45×10^{11} M⊙	1.7×10^{11} M⊙	**within 1.2×**

7%

THE MILKY WAY

the gap between the stars the feed delivers over 13 billion years and the stars our galaxy holds.

0

NUMBERS TUNED

to either galaxy — the same η and γ = 0.62, fixed by the early universe alone.

0.00002%

SAGITTARIUS A*

of the Milky Way's throat budget is held by its central black hole: 4×10^6 M⊙ against 2.1×10^{10}.

ONE FEED, FROM THE FIRST GALAXIES TO OUR OWN

The delivery law was fixed by JWST measurements at z ≈ 9 – 14.5 and never adjusted. Carried forward thirteen billion years, it builds the Milky Way and Andromeda to the masses they are measured to have. The same continuous throat feed reaches from the earliest galaxies to the galaxy we live in.

WHAT SUSTAINS THE PAIR

Not a collision question but a growth question: two structures fed without interruption for the age of the universe, each drawing from the 12 apex and 20 curvilinear throats, each ending with the stellar mass the feed predicts.

Halo and stellar masses from published Milky Way and Andromeda measurements; Sagittarius A at 4×10^6 M⊙. Feed integrated from z = 20 to z = 0 with the delivery law of page 15. Carried to z = 0 the feed builds more cosmic stellar mass than the present-day census records — the signature of a channel that never switches off.*

COMPUTED

Holding galaxies together is easy. Building them is the task.

53,000×

JADES-GS-Z14-0

light already emitted, against the power needed to bind the galaxy.

191,000×

JADES-GS-Z14-1

the same comparison — binding is never the limiting step.

106,000×

MOM-Z14

the same comparison for the most distant galaxy confirmed.

$$P_{bind} = (3/5)\ G\ M_{\star}^{2} / (R \cdot t_{age}) \quad \approx \quad 10^{31} - 10^{32}\ W$$

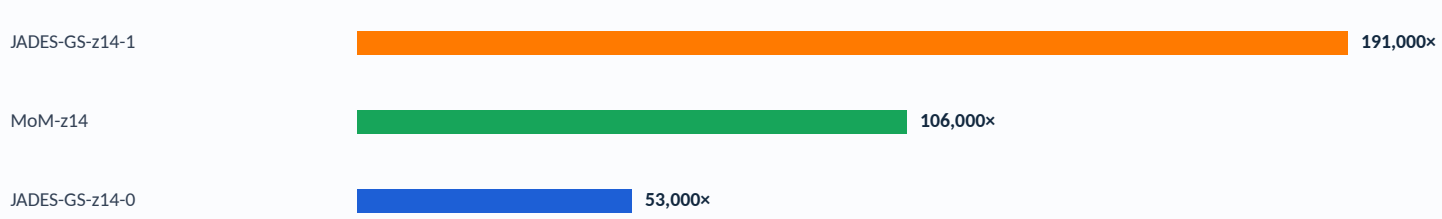
WHAT THIS IDENTIFIES

The earliest galaxies already radiate far more than their binding requires. The open problem is **mass assembly** — supplying the material itself, fast. That is exactly the job a continuous feed performs, and exactly where the throat channel is placed in this file.

WHY IT SHARPENS THE MODEL

It fixes what the throats must carry: not heat, not binding energy, but a steady supply of matter-energy at the rates on page 9. Every later page is built on that target.

LIGHT ALREADY EMITTED, AGAINST THE POWER NEEDED TO BIND THE GALAXY



Binding is never the limiting step.
The open problem is mass assembly —
the job of a continuous feed.

COMPUTED

The ceiling is about 660 times the need

6.6×10^8

ALLOWED · M☉ / MPC³

the most the feed can have delivered since z = 20, from page 7.

$\lesssim 10^6$

REQUIRED · M☉ / MPC³

the upper envelope of stellar mass density JWST infers at z ≈ 7–10.

≈ 660×

HEADROOM

room to spare between what the data allow and what the galaxies need.

THE STATEMENT THIS SUPPORTS

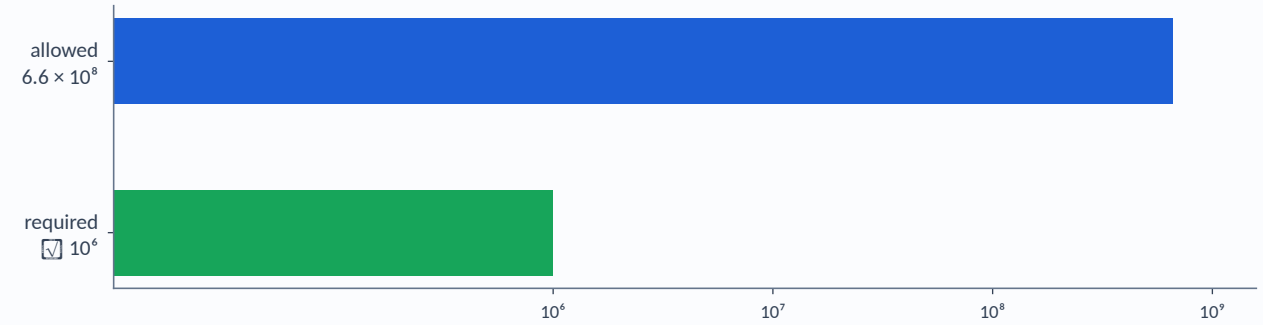
A continuous throat feed large enough to build every massive galaxy JWST has found in the early universe **fits inside** the combined Planck, DESI and Pantheon+ record — with nearly three orders of magnitude to spare.

WHY THE HEADROOM IS NATURAL

Stars at z ≈ 10 hold at most about one part in forty thousand of cosmic matter. The feed only has to be directed into structure; it never has to reshape the average universe to do its job.

Required envelope: the most massive JWST candidates at z ≈ 7–10. Allowed value: 1.65% × 4.0 × 10^{10} M☉ Mpc^{-3}.

ALLOWED AGAINST REQUIRED, MSUN PER MPC³ (LOG SCALE)



≈ 660×

headroom — nearly three orders of magnitude
Stars at z ≈ 10 hold ☑ 1 part in 40,000 of cosmic matter

THEOREM · FROM DECK 17

Identical throats, geometry-set delivery

1.43

APEX : CURVILINEAR

conductance to the hub, 9.487 against 6.635 — the same throat delivers differently by position.

2.879%

APEX-ORBIT CEILING

the largest share of any source flow a single edge can carry.

4.877%

CURVILINEAR CEILING

the same ceiling for the other orbit — two values, as symmetry requires.

THE FEED STAYS LOCAL

The transport localisation theorem proves the busiest edge is always the source's own throat, carrying exactly its effective resistance to the hub. Current decays at 0.40 per hop — a decay length of 1.1 hops. **What the throat feeds, it feeds nearby.** That is the umbilical cord in operator form.

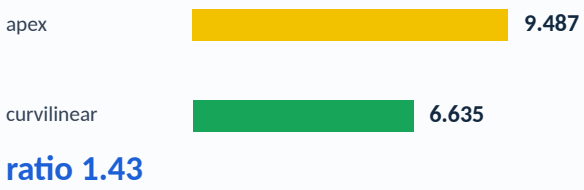
THE SPLIT IS FORCED, NOT CHOSEN

Every throat has the same coupling. The 43% difference in delivery comes entirely from the surface each throat sits in — 10 neighbours at an apex, 6 at a curvilinear. The architecture decides how the continuous feed is shared.

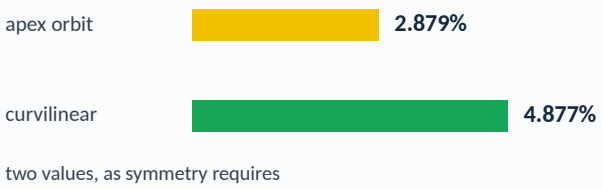
Results proved and computed in AI33-MPOPT · Geometric Transport Architecture (pages 4, 11 and 26), carried here unchanged.

HOW IDENTICAL THROATS SHARE THE FEED

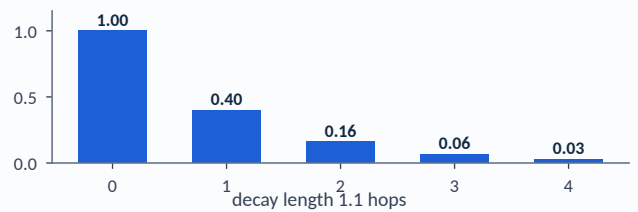
CONDUCTANCE TO THE HUB



LARGEST SHARE ONE EDGE CARRIES



CURRENT DECAY · 0.40 PER HOP



PROPOSAL

The reserve held by the hub

32

THE MAIN FEED

12 apex + 20 curvilinear throats carry the continuous supply after inception.

≈ 5%

THE RESERVE

Universe 33's own growth adds a helping hand — held available for structure when it is needed.

27 · 68 · 5

THE COSMIC BUDGET

dark matter · dark energy · ordinary matter, in percent — the partition the architecture is read against.

THE PICTURE

After inception the 32 throats feed the growing universe. The hub, Universe 33, grows with it and contributes a reserve of about five percent — not required to build the first galaxies, but present for finer structuring. The three-part budget of the universe is mirrored by three parts of the architecture: 12, 20 and the hub.

WHERE THE RESERVE FITS

A reserve directed into structure — into galaxies, not into the cosmic average — sits inside the ceiling of page 7 with room to spare. The headroom of page 21 is where it lives.

THE NEXT STEP FOR THIS PAGE

Derive the reserve fraction from the architecture itself — the throat share of total metric length is 25.93% and the domain shares are 41.14% and 32.93% — so that the five percent becomes a prediction of the geometry.

THE THREE-PART BUDGET AND THE THREE PARTS OF THE ARCHITECTURE



What, where, when and how

WHAT THE ARCHITECTURE FIXES	DATA COMPARED	WHEN	STATUS
Feed ceiling ≤ 1.65% of cosmic matter since z = 20	**DESI DR3 · Euclid · CMB-S4**	**2026 – 2030**	**established here**
Feed rates rising from 0.1 – 1.4 to ≈ 24 M☉ / yr, z ≈ 14 → 7	JWST spectroscopy across epochs	ongoing	target set here
Structure-following feed, η = 0.053%, fits cosmic star formation z = 9 – 14.5	**JWST PRIMER · JADES · NGDEEP**	**done**	**fits (page 11)**
The same feed, summed, fits the star-formation record z ≈ 9 → 6	**JWST CEERS · PRIMER · JADES**	**done**	**fits (page 12)**
Black holes first: black-hole share falls as the feed builds hosts	**JWST black-hole census, z ≈ 4 – 10.6**	**done**	**14 / 14 inside (page 14)**
Protocluster stellar mass from the concentrated throat feed	**JWST protocluster censuses, z ≈ 7.9 – 8.5**	**done**	**2 / 2 inside (page 16)**
Growth per expansion time Γ = sSFR / H(z) constant, z ≈ 5.4 – 14.2	**81 JWST galaxies across six samples**	**done**	**no trend (page 17)**
Capacity: highest JWST-inferred efficiencies against the throat-feed ceiling	**JWST efficiency measurements, z ≈ 6 – 10**	**done**	**3 / 3 inside, 4.4 – 29× headroom (page 18)**
Milky Way and Andromeda stellar mass from the same feed law	**published Local Group measurements**	**done**	**within 7% and 1.4× (page 19)**
Cosmic-web clustering 0.639 at matched density	**SDSS DR7 — observed 0.613**	**done**	**agrees (Deck 17, pp. 24–25)**
Transport-pessimal regime, Kirchhoff and Fiedler	SDSS DR7 galaxy network	done	same regime (Deck 17, pp. 24–25)
Degrees 11 and 7 with a hub of 32	halo catalogues · filament counts	next	prediction stated
Apex : curvilinear delivery 1.43	33-node circuit on a bench	anytime	prediction stated

WHY THIS ORDER

Each line pairs a number the architecture fixes in advance with a place it can be checked — nine already measured in agreement, one established here, three stated before their data exist.

A 33-node circuit makes the transport physical

152

RESISTORS

30 apex–apex · 60 apex–curvilinear · 30 curvilinear–curvilinear · 32 throats.

33

NODES

12 apex · 20 curvilinear · 1 hub. Every count fixed by the solid.

Exact

KIRCHHOFF

the operator is a conductance matrix, so a real circuit obeys it with no approximation.

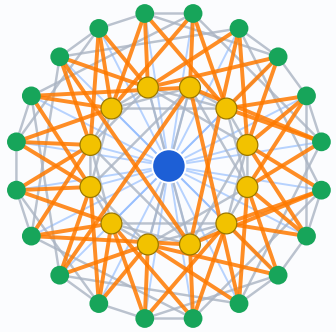
WHAT THE BENCH WILL SHOW

- Apex-to-hub conductance above curvilinear-to-hub by the geometry-set ratio of page 22.
- The busiest wire, for any injection point, is always that node's own throat.
- Current in equals current out at the hub; every other node balances to measurement precision.

WHY IT MATTERS

Every transport result in this collection is a theorem about a conductance network. A physical circuit turns those theorems into readings on a meter — the first time the architecture exists as a measurable physical object, in a laboratory, for the cost of a breadboard.

THE 33-NODE, 152-RESISTOR CIRCUIT



152 resistors: 30 + 60 + 30 + 32

33 nodes: 12 apex · 20 curvilinear · 1 hub

exact Kirchhoff: the operator is a conductance matrix

OPEN

Five predictions, written before their data

1 · THE CEILING HOLDS

As DESI DR3, Euclid and CMB-S4 arrive, the throat-feed fraction since z = 20 stays inside the 1.65% ceiling, and the ceiling tightens around the feed rather than excluding it.

2 · THE RATES ARE SUPPLIED

Galaxies measured at $z \gtrsim 14$ in future JWST programmes show assembly rates at or below the 0.1 – 1.4 $M_\odot$ / yr the throat channel carries at those epochs.

3 · THE BENCH READS 1.43

A 33-node, 152-resistor circuit with uniform throats returns an apex-to-curvilinear conductance ratio of 1.43 and localises every source current on its own throat.

4 · THE WEB, COMPARED AGAIN

A public halo catalogue, linked at the architecture's mean degree, returns a clustering coefficient near the architecture's 0.639. Compared with SDSS DR7: 0.613 observed against 0.639, four percent apart.

5 · THE FIRST 200 MILLION YEARS

Before any galaxy at z > 14.5 has a measured mass, the throat feed fixes the cosmic stellar mass density it can have built: **1.0×10^3 $M_\odot$ per Mpc³ at z = 15**, **7.5×10^1 at z = 18** and **1.2×10^1 at z = 20**, each with the ×3 census band of page 12. Recorded here for the confirmations to come.

WHY THEY ARE WRITTEN NOW

A prediction recorded before its data exists is the strongest statement a framework can make. These five are recorded here, dated, in the collection's final file.

FIVE PREDICTIONS, WRITTEN BEFORE THEIR DATA



SCOPE

The architecture meets the universe

ESTABLISHED HERE

- A continuous throat feed written as one physical equation with one strength parameter.
- A pipeline that recovers the standard universe from Planck, DESI and Pantheon+ jointly.
- Room for the feed: up to 1.65% of all cosmic matter since z = 20.
- A power target with units: about 10^{39} W per early massive galaxy, 10^{37}–10^{38} W per throat.
- The feed's task identified as mass assembly, with binding exceeded 53,000 to 191,000 times by emitted light.
- A growth history across four epochs, z ≈ 14 to z ≈ 7, with feed rates of the same order as measured star formation.
- A structure-following feed with one number, η = 0.053%, matching JWST cosmic star formation from z = 14.5 to z = 9, and carrying it forward to z ≈ 6 within 5%.
- An implosion–exit picture, and a black-hole-first ceiling that all 14 JWST black holes tested sit inside.
- A throat feed that grows with halo mass (γ = 0.62) and predicts the stellar mass of two JWST protoclusters.
- A dimensionless growth invariant, Γ = sSFR / H(z) ≈ 8, holding across z ≈ 5.4 – 14.2 and three decades of stellar mass.
- The same feed law, carried to z = 0, building the Milky Way to within 7% and Andromeda to within 1.4× of their measured stellar masses.
- Capacity headroom: about 660× against what JWST's galaxies require, and 4.4 – 29× even at the highest conversion efficiencies inferred.
- Five predictions recorded before their data.

NEXT TO ESTABLISH

- The feed amplitude η = 0.053% and the hub's five-percent reserve derived from the geometry itself.
- The feed traced galaxy by galaxy as new JWST measurements arrive.
- The 33-node circuit built and read.
- The halo-catalogue clustering measurement.
- The ceiling re-measured with DESI DR3 and Euclid.

Scope throughout: a one-parameter feed model fitted end to end to public data, and a finite architecture whose transport results are proved in the companion files.

MEASURED · COMPUTED

Every number, from public data

STEP	DATA AND METHOD	RESULT
Expansion history	DESI DR2 BAO — 13 measurements with published covariance	expansion likelihood
Relative distances	Pantheon+ — 1,590 supernovae, full statistical + systematic covariance	supernova likelihood, calibration marginalised
Recombination geometry	Planck 2018 compressed distance priors	$R = 1.7502 \cdot \ell_A = 301.471 \cdot \Omega_b h^2 = 0.02236$
Sound horizon · recombination	standard fitting formulae	r_d, $z_\star$, r_s computed per model
Joint cosmological fit	all three data sets fitted together	$h = 0.687 \cdot \Omega_m = 0.306 \cdot 1.65\%$ ceiling
Galaxy power targets	JWST masses, radii and UV luminosities	pages 8 – 9, 20 – 21
Structure-following feed	JWST star-formation density · standard halo and matter-spectrum models	page 11 · $\eta = 0.053\%$, $\chi^2 = 1.3$
Feed summed into stellar mass	JWST stellar-mass function · standard IMF conversions	page 12 · 1.00 – 1.05, $z \approx 9 \rightarrow 6$
Black holes first	JWST black-hole and host masses · feed and observed growth histories	page 14 · 14 / 14 inside
Concentrated throat feed	JWST stellar-mass function · protocluster censuses · standard structure growth	pages 15 – 16 · $\gamma = 0.62$ · 2 / 2 inside
Growth-to-expansion ratio	JWST stellar masses and star-formation rates · H(z) from the joint fit	page 17 · $\Gamma = 8.0$, no redshift trend
Capacity benchmark	JWST-inferred conversion efficiencies · the ceiling of page 7	page 18 · 3.4 – 22% of capacity used
Local Group benchmark	published Milky Way and Andromeda masses · feed integrated to z = 0	page 19 · within 7% and 1.4×

THE SINGLE SENTENCE THIS FILE SUPPORTS

The 33-domain architecture carries a continuous throat feed that the modern cosmological record has room for — enough to build the earliest massive galaxies, to keep pace with them as they grew, and to reach the Milky Way itself.

Companion files: 13 Quantum-Gravity Bridge · 14 Loop Quantum Gravity Bridge · 15 Lawfulness · 16 Super Formula Build-Up · 17 Geometric Transport Architecture · 19 Cosmic Web Atlas · 20 The Rivero-Zeta Curve · 21 The 33-Multiverse Map.